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# ANALYSIS OF DATA COMMUNICATION IN WEB APPLICATION USING BROWSER DEVELOPER TOOLS

## 1. AIM

To analyze and understand the communication of data between a web browser and a web application by using Browser Developer Tools. The study focuses on observing HTTP/HTTPS requests and responses, identifying the data exchanged between the client and server, and understanding how web applications transfer resources such as HTML, CSS, JavaScript, images, form data, and API responses.

The main purpose of this study is to provide practical knowledge about what happens behind a web page when a user performs an action such as opening a website, submitting a form, clicking a button, logging in, or requesting information. Browser Developer Tools provide a convenient environment for inspecting these communication processes without requiring specialized network-monitoring software.

## 2. ABSTRACT

Web applications depend on continuous communication between the client side and the server side. When a user visits a web application, the browser sends requests to a web server and receives responses containing the resources or information required to display and operate the application. This communication commonly takes place through HTTP or HTTPS protocols. Although these activities happen automatically in the background, Browser Developer Tools make the process visible and easier to study.

This study presents an analysis of data communication in a web application using the Developer Tools available in modern browsers. The Network panel is used to observe requests generated while loading a web page and while interacting with application features. Important details such as request method, URL, status code, response type, request headers, response headers, payload, timing information, and transferred data can be examined. These observations help explain the relationship between a browser and a web server.

The analysis also demonstrates how different HTTP methods, including GET and POST, are used for different operations. A GET request is generally used to retrieve resources or

information, whereas a POST request can be used to send data to the server. Status codes such as 200, 201, 301, 304, 400, 401, 403, 404, and 500 provide information about the result of a request. By examining these values, students can develop a practical understanding of successful communication, redirection, client-side errors, authorization problems, and server-side failures.

The study further considers the role of HTTPS in protecting communication between the browser and server. While Developer Tools can display request and response information available to the browser, encrypted HTTPS traffic is protected during transmission by TLS. Therefore, the analysis helps students understand both the observable structure of web communication and the importance of secure data transfer.

## 3. INTRODUCTION

A web application is a software system that operates through a web browser and communicates with one or more servers. Examples include online banking systems, e-commerce websites, social networking platforms, student portals, and online reservation systems. In these applications, data communication is essential because the browser must request information from servers and send user-generated information back to them.

When a user enters a web address, the browser begins a sequence of communication activities. It may resolve the domain name, establish a secure connection, send an HTTP request, receive an HTTP response, and download additional resources. Modern applications may also communicate with application programming interfaces (APIs) to retrieve or submit structured data, often in JSON format. These requests can occur repeatedly while a page is being used.

Browser Developer Tools provide built-in features for examining this process. The Network panel is particularly useful because it records network activity and presents information about each request. A student can reload a web page, select an individual request, and inspect its headers, payload, response, and timing. This makes an otherwise invisible communication process observable and easier to understand.

## 4. OBJECTIVES

- To understand the basic concept of client-server communication in web applications.
- To identify HTTP/HTTPS requests and responses generated by a browser.
- To examine common HTTP methods such as GET and POST.
- To study HTTP response status codes and their meanings.
- To inspect request and response headers, payloads, and response data.
- To understand how browser interactions generate network requests.
- To observe API communication and structured data such as JSON.

- To understand the importance of HTTPS and secure data communication.

## 5. SCOPE OF THE STUDY

The study is limited to observing data communication from the browser perspective using built-in Browser Developer Tools. It covers network requests, responses, headers, payloads, status codes, transferred resources, and basic timing information. The study does not attempt to decrypt protected HTTPS traffic or access confidential information. It is intended for educational analysis using publicly accessible or authorized web applications.

## 6. EXPECTED OUTCOME

At the end of the study, the learner should be able to open Developer Tools, use the Network panel, identify requests generated by a web page, interpret important HTTP information, and explain how data moves between a browser and a server. The learner should also be able to distinguish successful requests from common client and server errors and describe the role of HTTPS in protecting communication.